

INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 5: Discussion

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5 Discussion

This chapter interprets the Chapter 4 findings against RQ1–RQ4 and SRQ1, discusses theoretical and practical implications for DeFi risk assessment, formulates a Sybil attack cost model on allowance graphs, addresses regulatory and ethical considerations, evaluates threats to validity, and states limitations of the primary EndorseRank–AWP comparison. Throughout, empirical quantities are those reported for the matched wallet cohort ($n = 5,521$) unless otherwise noted: inter-method Kendall $\tau = 0.316$, EndorseRank allowance $\tau = 0.355$, AWP transfer $\tau = 0.534$, trading-success $\tau = 0.096$ (EndorseRank) and 0.179 (AWP), GF-PR trading-success $\tau = 0.583$ and inverse-risk $\tau = 0.376$, and PageRank runtimes of 2.105 s versus 18.711 s ($\approx 8.9\times$), with pool scaling of $\approx 10.4\times$ at $n = 10,000$ and $\approx 9.4\times$ at $N = 31,612$.

5.1 Findings by research question

The primary empirical comparison concerns EndorseRank and AWP as social reputation methods on wallet-to-wallet graphs. The following subsections interpret each research question in light of the five-proxy alignment design and the computational benchmarks.

5.1.1 RQ1: Reputation structure and rank divergence

RQ1 asks whether EndorseRank and AWP produce meaningfully different wallet orderings on an identical Arbitrum sample. Inter-method Kendall τ between the two score vectors is 0.316 on the matched cohort ($n = 5,521$), within the weak-to-moderate alignment band defined in Chapter 2. Correlation is present—both methods operate on overlapping wallet activity in the same observation window—but the magnitude is far from unity. Allowance and transfer graphs therefore emphasize different wallet relationships.

This divergence is structurally expected. The endorsement graph G_{approve} contains 14,727 edges encoding current spending authorization; the transfer graph G_{transfer} contains 137,087 edges encoding time-decayed economic flow ^[16]. PageRank propagates influence along these edges under the same damping convention ^{[40],[49]}, so differences in edge sets and weights produce differences in stationary mass. Allowances encode authorization intent; transfers encode movement of value. A wallet that receives large transfers without granting allowances can rank highly under AWP and modestly under EndorseRank, and conversely for a wallet that authorizes spenders without being a transfer hub.

H1 is therefore supported: EndorseRank and AWP yield partially overlapping yet construct-distinct orderings. The practical implication is that the two scores are not interchangeable substitutes. Deployments that adopt EndorseRank for efficiency must accept a different ranking geometry, not merely a faster approximation of AWP.

5.1.2 RQ2: Construct-specific social-method alignment

RQ2 asks which social method aligns more strongly with each of the five proxy families. Table 4.7 and the per-family tables (Tables 4.8–4.12) show clear construct-specific winners rather than a universal ranking.

On the **allowance** family, EndorseRank achieves mean Kendall $\tau = 0.355$, whereas AWP is near zero (-0.030). This pattern is the strongest evidence that EndorseRank measures its intended construct: latest-allowance centrality aligns with allowance-based proxies, while transfer-graph PageRank does not. In measurement terms, the indicator tracks the construct it was designed to represent ^[14].

On the **transfer** family, AWP achieves $\tau = 0.534$ and EndorseRank a moderate secondary-axis $\tau = 0.333$. AWP’s advantage is expected because transfer proxies (in-degree and in-value) are the validation frame of Do et al.^[16] and share input semantics with G_{transfer} . EndorseRank’s moderate transfer alignment indicates partial overlap—wallets active in approvals often participate in transfers—but not equivalence.

On **trading-success** proxies derived from GMX V2 perpetual closes, AWP exceeds EndorseRank among social methods ($\tau = 0.179$ vs. 0.096). Allowance-based propagation does not align more strongly with trading-success proxies than transfer-based AWP on this cohort. EndorseRank’s $\tau = 0.096$ falls in the negligible-noise band and AWP’s $\tau = 0.179$ in the weak-to-moderate band; overall, social-method trading-success alignment remains weak, and neither method approaches the outcome-native GF-PR ceiling discussed under SRQ1. On **inverse-risk** and **Sybil-adjusted stability** families, AWP likewise leads among social methods, consistent with transfer activity capturing more of the variance in leveraged outcomes and activity-adjusted stability than allowance snapshots alone.

H2 is supported in its construct-specific form: EndorseRank wins on allowance; AWP wins on transfer and most social-outcome families. The result argues against treating reputation as a single latent trait measured equally well by any graph. Instead, graph semantics determine which external proxies a score can credibly track.

5.1.3 RQ3: Computational efficiency

RQ3 asks whether EndorseRank is computationally more efficient than AWP on the same matched sample. On the full cohort ($n = 5,521$), EndorseRank completes PageRank in 2.105 s versus 18.711 s for AWP—approximately $8.9\times$ faster—with a sparser graph (14,727 vs. 137,087 edges). Both methods converge within tolerance 10^{-8} ; EndorseRank requires fewer iterations on average, consistent with a smaller effective edge set ^[49].

Scaling on the expanded wallet pool ($N = 31,612$) preserves the advantage: $\approx 10.4\times$ at $n = 10,000$ and $\approx 9.4\times$ at full pool size. Robustness checks on damping and top-token subgraphs do not alter the allowance–transfer trade-off pattern: EndorseRank’s allowance τ remains stable near 0.355 across damping $d \in \{0.75, 0.85, 0.95\}$, while transfer-family τ for both methods varies by at most approximately 0.005.

H3 is supported. The efficiency gain follows from latest-allowance snapshots instead of aggregating full transfer history with logistic decay ^[16]. For protocols that must refresh reputation frequently,

this difference is operationally material: an order-of-magnitude reduction in PageRank wall time enables tighter update loops without proportional infrastructure cost.

5.1.4 RQ4: Transfer versus trading-success validation frames

RQ4 asks which validation frame yields stronger rank alignment for social reputation methods on this chain: Do et al. transfer proxies or GMX V2 trading-success proxies. Transfer proxies yield higher mean τ than trading success for both EndorseRank and AWP (Table 4.7). AWP’s transfer $\tau = 0.534$ substantially exceeds its trading-success $\tau = 0.179$; EndorseRank’s transfer $\tau = 0.333$ exceeds its trading-success $\tau = 0.096$.

Two interpretations follow. First, social graphs built from wallet-to-wallet economic relationships are more tightly coupled to transfer centrality than to perpetual trading profitability. Trading success depends on market timing, leverage choice, and liquidation dynamics that are only partially reflected in approval or transfer topology ^{[23],[44]}. Second, outcome-based proxies show weak-to-modest alignment overall for social graphs, suggesting that no single proxy family fully captures “reputation” on this sample. Domain alignment remains a relative, multi-construct exercise ^{[14],[36],[54]}, not a search for one ground-truth ranking.

RQ4 therefore favors transfer proxies as the stronger validation frame for social methods on Arbitrum One in this window, while leaving room for trading-success proxies as a secondary, economically motivated axis—especially for protocols whose risk is defined by leveraged outcomes rather than transfer hubs.

5.2 Supplementary finding (SRQ1)

GF-PR achieves the highest trading-success mean τ (0.583) and inverse-risk mean τ (0.376; Table 4.7), as expected for an outcome-native graph built from the same PnL stream used to define those proxies. The POOL/SINK star construction ranks wallets by synthetic flow of realized gains and losses; it shares input semantics with trading-success and inverse-risk proxies by design. High

alignment is therefore evidence of *construct overlap*, not of superior social-graph semantics ^[14].

This ceiling clarifies interpretation of social-method results. EndorseRank’s trading-success $\tau = 0.096$ and AWP’s $\tau = 0.179$ should be read against an outcome-native upper bound of 0.583, not against an implicit expectation of near-perfect prediction. Among deployable social methods, AWP still exceeds EndorseRank on trading-success proxies, confirming that transfer-based propagation aligns more strongly with trading outcomes on this cohort than allowance-based propagation, independent of the GF-PR ceiling. EndorseRank and AWP remain the deployable social reputation methods; GF-PR is a diagnostic reference only (Claim SC1).

5.3 Theoretical implications

The findings extend three strands of theory: hyperlink-style propagation on financial graphs, construct validity for on-chain scores, and the economics of trust under pseudonymity.

5.3.1 Propagation semantics beyond transfers

PageRank’s original formulation treats inbound hyperlinks as citations ^{[8],[49]}. Do et al.^[16] map that metaphor to Ethereum transfers. EndorseRank maps it to ERC-20 allowances: an owner who authorizes a spender issues a directed endorsement of spending authority ^{[18],[19]}. The inter-method $\tau = 0.316$ demonstrates that changing the edge semantics changes the ranking geometry even when the node set is held fixed. Graph-based reputation is therefore not a single algorithm applied to “the” transaction graph; it is a family of methods whose validity depends on which relationships are encoded as edges ^{[40],[44]}.

Related trust-propagation systems in peer-to-peer and web settings—EigenTrust and TrustRank—similarly show that seed sets and edge definitions dominate outcomes ^{[27],[35]}. EndorseRank contributes an edge definition native to token standards rather than to message-passing overlays or hyperlink graphs.

5.3.2 Multi-construct domain alignment

The five-proxy framework extends Do et al.’s validation logic by separating transfer centrality, allowance delegation, leveraged outcomes, and Sybil-adjusted activity. EndorseRank aligns with its primary allowance construct (G_{approve} , $\tau = 0.355$); transfer proxies serve as a secondary validation axis (moderate $\tau = 0.333$, below AWP 0.534). AWP aligns with transfer—its primary graph construct—and several outcome proxies derived from the same wallet activity. The supplementary GF-PR analysis demonstrates the upper bound of trading-outcome proxy alignment achievable when ranking directly on outcome flow rather than social edges (SRQ1).

This pattern is consistent with classical construct-validity reasoning: indicators should correlate with measures of the same construct and diverge from measures of different constructs ^[14]. Rank correlations ^{[36],[54]} operationalize that reasoning for ordinal reputation scores without imposing classification thresholds.

5.3.3 Reputation under cheap pseudonyms

Public ledgers permit cheap identity creation ^{[17],[46],[60]}. Reputation systems that reward inbound transfers or allowances are therefore exposed to Sybil strategies in which an attacker manufactures edges among controlled addresses. The empirical results do not eliminate that threat; they show that allowance and transfer graphs respond differently to the activity patterns present in the observational sample. Section 5.5 formalizes attack cost on allowance graphs, arguing that non-zero allowances to external spenders can impose opportunity and revocation costs that pure transfer wash trading does not.

5.4 Practical implications and deployment scenarios

Protocols and infrastructure providers face a design choice between transfer-hub detection and authorization-aware reputation. The empirical trade-off supports scenario-dependent adoption rather than a single recommended score.

5.4.1 Transfer-hub and flow monitoring

Protocols that prioritize detection of high-flow wallets—for example, for market-maker identification, routing analysis, or transfer-based compliance heuristics—may prefer AWP despite higher cost. AWP’s transfer $\tau = 0.534$ and stronger social-outcome alignment make it the better match when the operational target is economic flow centrality^[16]. The runtime of 18.711 s on $n = 5,521$ wallets is acceptable for batch analytics; it is less attractive for sub-second or near-real-time refresh.

5.4.2 Authorization-aware and low-latency refresh

Protocols that emphasize explicit spending authorization—token vaults, allowance-gated integrations, or risk engines that treat approvals as trust acts—may prefer EndorseRank. The method’s allowance $\tau = 0.355$, sparse graph (14,727 edges), and 2.105 s PageRank time support frequent recomputation. Pool scaling ($\approx 10.4\times$ at $n = 10,000$; $\approx 9.4\times$ at $N = 31,612$) indicates that the efficiency advantage persists beyond the matched cohort. Operators accept lower transfer-proxy alignment ($\tau = 0.333$) and weaker trading-success alignment ($\tau = 0.096$) in exchange for authorization semantics and speed.

5.4.3 Evaluation design and hybrid use

GF-PR is suitable only as a diagnostic ceiling for outcome-proxy overlap during evaluation design, not as a deployable social reputation score. Evaluation pipelines should report social-method τ alongside an outcome-native reference when trading-success proxies are used, to avoid mistaking construct overlap for social validity (Claim SC1).

Hybrid deployments are feasible: EndorseRank for continuous authorization-aware monitoring, AWP for periodic transfer-hub audits, and GMX-linked outcome metrics as separate risk dashboards rather than as substitutes for social scores^{[23],[53]}. The open pipeline enables cross-protocol replication on any ERC-20 chain with allowance logs (Appendix 8.1). Integration with live risk oracles and governance tooling remains future work (Chapter 6), but the efficiency profile of En-

dorseRank is compatible with oracle update cadences that transfer-graph AWP may not sustain at equal cost.

5.4.4 Industry adoption scenarios

The practical value of EndorseRank and AWP is not that either score predicts trading profit—social-method trading-success alignment is weak overall ($\tau = 0.096$ negligible for EndorseRank; $\tau = 0.179$ weak-to-moderate for AWP)—but that both supply a cheap, standardized, and independently recomputable ordinal signal that protocols can plug into operational decisions. Value capture is therefore hypothesized to run through fees, utilization, monitoring cost, and reduced reward leakage, rather than through reducing collateral or transferring protocol credit risk to the reputation score. The scenarios below map actors to decision problems, method choices, and revenue mechanisms consistent with the empirical results.

- **Perpetual DEX operators:** Protocols such as GMX V2 earn primarily from trading fees tied to notional volume, not from trader PnL ^[23]. Operators face a continuous decision problem: which wallets to prioritize for monitoring, fee tiers, or feature access when reputation must be refreshed frequently. EndorseRank fits this loop because allowance $\tau = 0.355$ aligns the score with authorization activity and PageRank completes in 2.105 s on the matched cohort, enabling frequent (order-of-seconds) updates without transfer-history aggregation. Any revenue effect would be hypothesized to run through fee volume (if differentiated service raises retained notional) and lower monitoring cost of risk review—not through selecting profitable traders. Fee and monitoring policies remain the justified levers.
- **Lending protocols:** Lending markets seek higher capital utilization without abandoning collateralization ^{[5],[26]}. Reputation can enter as an auxiliary signal for interest-rate or fee tiers and review cadence on already-collateralized positions, not as a substitute for collateral. EndorseRank is the natural social input when the protocol already relies on ERC-20 approvals for deposits and borrows (Section 2.7). Utilization-driven interest income is the revenue channel; the weak trading-success alignment of social methods rules out treating the score as a default

predictor. Any collateral-differentiation experiment remains future work (Chapter 6).

- **Wallet and security tooling:** Users and integrators must decide whether a spender contract is widely authorized. Displaying the spender’s EndorseRank—or the distribution of inbound endorsement mass—as a trust badge operationalizes allowance semantics as explicit delegation of spending authority ^{[18],[19]}. Vendors may monetize through premium alerts and API subscriptions; this is subscription revenue for tooling, distinct from protocol fees, utilization, monitoring-cost savings, or Sybil-filter reward-leakage reduction. The badge reports graph position under published edge definitions; it is not a security audit or a guarantee against malicious contracts.
- **Liquidity providers and market makers:** LPs and market makers need to identify transfer hubs and flow structure for inventory and spread management. AWP is the better social method for this target: transfer $\tau = 0.534$ and a batch-friendly runtime of 18.711 s on $n = 5,521$ wallets support periodic analytics rather than sub-second refresh ^[16]. The operational channel is lower screening and monitoring cost when identifying transfer hubs—not improved P&L or a profitability forecast. AWP remains a flow-centrality instrument, not a profitability oracle.
- **Airdrop and governance programs:** Token distributions and governance eligibility are routinely gamed by Sybil clusters ^[17]. Programs can require inbound endorsement diversity or apply the closed-cluster mass bound of Section 5.5 (Equation 5.8) as a filter before rewards are paid. The revenue—or cost-avoidance—channel is reduced reward leakage to fabricated identities. Filters based on graph structure mitigate closed-cluster inflation; they do not eliminate open-cluster attacks that recruit honest inbound endorsements.
- **Analytics and compliance providers:** Exchanges, funds, and compliance teams need auditable counterparty rankings that third parties can recompute. The open pipeline (Appendix 8.1) supports subscription APIs that publish EndorseRank and AWP under disclosed windows and parameters. Recurring data revenue is a vendor business model; auditability is

the product differentiator relative to opaque machine-learning scores ^[48]. Providers should disclose weak trading-success alignment and Sybil exposure alongside any dashboard that surfaces ranks.

None of these scenarios treats reputation as a substitute for collateral or as a predictor of trading profit. The claim is narrower and, we argue, more durable: a standardized, auditable, low-latency ordinal signal lowers the cost of counterparty differentiation, and the protocols that adopt it may capture value through fees, utilization, monitoring cost, and reduced reward leakage rather than through reducing collateral or transferring protocol credit risk to the score.

5.4.5 From valuation multiples to reputation multiples

When price–earnings and price–book ratios were first popularized, their value did not come from predictive accuracy—they are imperfect return predictors—but from standardization: any investor could compute the same number from public filings, compare firms on a common scale, and communicate risk appetite in a shared vocabulary ^{[6],[20],[24]}. Token-economy valuation models similarly rely on shared, publicly observable quantities rather than private identity ^[13]. On-chain reputation scores are positioned to play an analogous role for pseudonymous wallets. EndorseRank and AWP are computed from public logs by an open pipeline (Appendix 8.1), are cheap enough to refresh continuously (2.105 s on the matched cohort for EndorseRank), and induce orderings that any counterparty can independently verify.

Standardization is the first link in the analogy. Accounting conventions made price–earnings ratios computable from disclosed statements ^[24]; ERC-20 event standards make allowance- and transfer-based ranks computable from broadcast logs ^{[18],[19]}. Just as GAAP-like reporting enabled a common valuation language, token-log standards enable a common reputation language without requiring off-chain identity.

Comparability is the second link. Pseudonymous markets suffer from information asymmetry: counterparties cannot cheaply distinguish authorization-active wallets from inactive or fabricated ones ^[1]. A shared ordinal score reduces that asymmetry by giving protocols, LPs, and users a

common scale on which to screen and communicate. Lower screening costs may support broader participation even when the score is only a coarse filter; liquidity effects are not estimated here.

Imperfection is the third link, and it is a feature of the argument rather than a concession. Empirical finance has long documented that valuation multiples are noisy predictors of subsequent returns [6],[20]; their institutional role is screening, communication, and process design. Likewise, the weak trading-success alignment of EndorseRank and AWP ($\tau = 0.096$ and 0.179) rules out reading either score as a profit forecast, but does not erase their value as construct-specific ordinal instruments. Quantifying that weakness is itself part of responsible adoption: operators know which decisions the scores can and cannot support.

Gaming is the fourth link. Earnings management shows that once a metric becomes decision-relevant, agents invest in distorting it [31]. Allowance-based reputation invites approval farming and Sybil clusters in the same spirit. Section 5.5 formalizes attack cost on endorsement graphs, and Section 5.6 argues for disclosure of graph definitions, observation windows, and known failure modes. Standardized reputation scores, like valuation multiples, are useful only when their construction is public and their failure modes are named.

The ecosystem effect follows from these four links. A shared, auditable reputation vocabulary can lower the cost of counterparty differentiation across perpetual markets, lending pools, wallets, and compliance tooling, channeling protocol revenue through fees, utilization, monitoring cost, and reduced reward leakage while preserving collateral as the primary risk backstop. Adoption pathways that overclaim predictive power for loan default or trading profit would undermine that vocabulary; pathways that treat scores as standardized ordinal infrastructure are consistent with the evidence in this dissertation.

5.5 Sybil attack cost model on allowance graphs

Sybil attacks exploit cheap pseudonyms to manufacture reputation [17]. On transfer graphs, an attacker can wash-trade value among controlled wallets, inflating in-degree and in-value at the cost

of fees and inventory risk. On allowance graphs, the attack surface differs because edges represent authorization rather than completed transfers. This section formalizes a minimal cost model for inflating EndorseRank via fabricated allowances.

5.5.1 Attack setup

Let $\mathcal{A}$ be the set of addresses controlled by an attacker and $\mathcal{H}$ the set of honest wallets in the endorsement graph $G_{\text{approve}} = (V, E, w)$. An edge (u, v) with weight $w_{uv} > 0$ denotes that u currently authorizes v to spend decimal-adjusted token units totaling w_{uv} . EndorseRank computes the PageRank vector $\mathbf{r}$ on G_{approve} with damping $d \in (0, 1)$ ^{[40],[49]}:

$$\mathbf{r} = d P^\top \mathbf{r} + \frac{1-d}{|V|} \mathbf{1}, \quad (5.1)$$

where P is the row-stochastic matrix obtained by normalizing outgoing allowance weights (with a standard handling of dangling nodes).

The attacker seeks to maximize the mean EndorseRank of a target subset $T \subseteq \mathcal{A}$,

$$J(T) = \frac{1}{|T|} \sum_{i \in T} r_i, \quad (5.2)$$

subject to a budget on attack cost defined below.

5.5.2 Cost components

Fabricating an allowance edge (u, v) with weight w_{uv} incurs at least three cost components:

1. **Transaction cost:** Submitting `approve` or `permit` consumes gas. Let c_{gas} be the expected fee per allowance update. For m updates,

$$C_{\text{gas}} = m c_{\text{gas}}. \quad (5.3)$$

2. **Capital lock / opportunity cost:** A non-zero allowance to an external spender $v \notin \mathcal{A}$ exposes

tokens owned by u to transfer by v . Even if $v \in \mathcal{A}$, maintaining large allowances to contracts or routers may be required to mimic honest behavior. Let $\ell(w_{uv})$ be the opportunity cost of capital associated with weight w_{uv} over the reputation refresh horizon (for example, forgone yield). Then

$$C_{\text{cap}} = \sum_{(u,v) \in E_{\mathcal{A}}} \ell(w_{uv}), \quad (5.4)$$

where $E_{\mathcal{A}}$ denotes attacker-created edges.

3. **Revocation and detection cost:** Honest users revoke stale allowances; attackers who never revoke may be distinguishable. Let c_{rev} be the expected cost of periodic revocation and re-approval needed to maintain a target edge set over k refresh epochs:

$$C_{\text{rev}} = k c_{\text{rev}}. \quad (5.5)$$

Total attack cost is

$$C = C_{\text{gas}} + C_{\text{cap}} + C_{\text{rev}}. \quad (5.6)$$

5.5.3 Influence and a cost-effectiveness bound

Under standard PageRank sensitivity results, the influence of an edge on $\mathbf{r}$ is mediated by the stationary mass at the source and the normalized outgoing weight^[40]. A coarse upper bound on the gain in $J(T)$ from adding edges among $\mathcal{A}$ alone—a closed Sybil cluster with no honest inbound endorsements—is limited by the teleportation mass:

$$\sum_{i \in \mathcal{A}} r_i \leq |\mathcal{A}| \cdot \frac{1-d}{|V|} + d \cdot \sum_{i \in \mathcal{A}} r_i, \quad (5.7)$$

which rearranges to

$$\sum_{i \in \mathcal{A}} r_i \leq \frac{|\mathcal{A}|}{|V|}. \quad (5.8)$$

Thus a closed attacker cluster cannot capture more than its population share of total rank mass. To exceed that share, the attacker needs inbound endorsements from $\mathcal{H}$, which requires either compromising honest wallets or inducing honest owners to approve attacker-controlled spenders—actions whose cost is typically dominated by C_{cap} and social-engineering effort rather than gas alone.

When the attacker purchases inbound allowances from honest owners (for example, by operating a service that requests approvals), the marginal cost per unit of honest-sourced weight w_{ha} can be written

$$c_{\text{in}}(w_{ha}) = c_{\text{gas}} + \ell_{\text{honest}}(w_{ha}) + \pi(w_{ha}), \quad (5.9)$$

where ℓ_{honest} is compensation paid to the honest owner and π is a risk premium for potential loss from malicious spending. Cost-effectiveness of the attack is then

$$\eta = \frac{\Delta J(T)}{C}, \quad (5.10)$$

with $\Delta J(T)$ the increase in mean target rank attributable to the attack edges.

Equations (5.7)–(5.10) imply that EndorseRank is not Sybil-proof, but that closed-cluster inflation is population-share limited, and that cross-cluster inflation requires costly honest-sourced allowances. Transfer wash trading faces different costs (fees and inventory) but does not require exposing spending authority to external addresses. Sybil-adjusted stability proxies in Chapter 4 partially address observational manipulation; they do not replace adversarial evaluation, which remains future work.

5.6 Regulatory and ethical considerations

On-chain reputation scores occupy a contested space between transparency and fairness. Legal and regulatory analyses of DeFi emphasize that decentralized markets remain subject to disclosure and consumer-protection concerns even when settlement is on-chain^[62]. Packin and Lev-Aretz^[48] caution that decentralized credit scoring can reproduce black-box harms if models are opaque, un-

appealable, or coupled to identity systems without due process. EndorseRank and AWP are fully determined by public ledger events and published algorithms; any party can recompute scores from the open pipeline (Appendix 8.1). That auditability is a partial response to black-box concerns, but it does not resolve distributional effects: wallets with limited approval or transfer history receive low scores regardless of off-chain creditworthiness.

Ethically, scores must not be presented as identity, creditworthiness in the consumer-lending sense, or proof of unique personhood^[17]. They are ordinal indicators of on-chain graph position under specified edge semantics. Deployments that gate access to financial services should disclose the graph definition, observation window, and known failure modes—including Sybil exposure and weak trading-success alignment for social methods ($\tau = 0.096$ and 0.179). Regulatory regimes that require explainability favor graph propagation over opaque machine-learning scores^{[39],[48]}, provided operators resist overclaiming predictive power for loan default.

Privacy is structurally limited on public ledgers^{[46],[60]}: reputation computation uses data already broadcast. The ethical obligation is therefore not to avoid analysis but to avoid deanonymization beyond address-level aggregation and to prevent misuse of scores as persistent biometric-like identifiers. This dissertation analyzes addresses only; no off-chain identity linkage is performed (Chapter 3).

5.7 Threats to validity

Interpretation of the empirical claims should account for the following threats, organized by construct, internal, external, and adversarial validity.

5.7.1 Construct validity

- **Proxy not ground truth:** No under-collateralized lending default labels are available at sufficient density. Perpetual-trading liquidation and PnL on GMX V2 are domain-intuition proxies, not loan repayment^{[23],[44]}. Alignment with trading-success proxies ($\tau = 0.096$ for En-

dorseRank; 0.179 for AWP) must not be read as credit-default prediction.

- **GF-PR construct overlap:** GF-PR shares input semantics with trading-success and inverse-risk proxies; its high τ on those families (0.583 and 0.376) must not be interpreted as superior social reputation ^[14].
- **Secondary-axis interpretation:** EndorseRank’s moderate transfer $\tau = 0.333$ is a secondary validation axis, not evidence that allowance graphs measure transfer centrality as well as AWP ($\tau = 0.534$).

5.7.2 Internal validity

- **Shared observation window:** All methods and proxies are computed over 2025-12-01 to 2026-05-31. Market regimes specific to that window—volatility, incentive programs, or protocol changes—may inflate or deflate alignment. Robustness checks on damping and top-token subgraphs reduce sensitivity to PageRank hyperparameters and token concentration but do not vary the calendar window.
- **Cohort selection:** The matched wallet cohort requires at least three GMX closes and non-zero reputation subgraph connectivity ($n = 5,521$; 365,488 `PositionDecrease` events). Results may not extend to wallets with thinner perpetual histories or to addresses absent from allowance and transfer subgraphs.
- **Edge-weight measurement:** Token-decimal normalization, USD conversion choices, and permit-versus-approve coverage can affect w_{uv} . Measurement error in edge weights propagates to $\mathbf{r}$ and to rank correlations ^[40].

5.7.3 External validity

- **Single protocol and chain:** Evidence is limited to Arbitrum One and GMX V2. Generalization to other perpetual markets, lending protocols, or Layer-1 chains requires cross-protocol validation ^{[28],[53],[59]}.
- **Pool scaling versus alignment scaling:** Runtime scaling on the expanded wallet pool ($N =$

31,612) supports efficiency claims at larger n ; alignment results remain anchored to the matched cohort with GMX outcomes. Efficiency gains should not be assumed to preserve the same τ profile at arbitrary population sizes without further alignment studies.

5.7.4 Adversarial validity

- **Sybil and wash trading:** Adversarial allowance or transfer patterns may manipulate rankings^[17]. Sybil-adjusted proxies partially address but do not eliminate this risk. The cost model in Section 5.5 is analytical; it is not an empirical red-team evaluation.
- **Strategic approval behavior:** Once EndorseRank is deployed, rational actors may grant allowances strategically to farm rank. Observational τ estimated before widespread deployment may overstate robustness to strategic behavior.

5.8 Limitations

Results reflect one six-month window on Arbitrum One for a matched wallet cohort of $n = 5,521$ addresses. The endorsement subgraph comprises 14,727 edges; the transfer subgraph comprises 137,087 edges. Trading-success alignment for social methods is weak to modest (0.096 and 0.179), and transfer proxies remain the stronger validation frame (RQ4). EndorseRank’s primary demonstrated strengths are allowance-construct alignment ($\tau = 0.355$), rank distinctness from AWP (inter-method $\tau = 0.316$), and computational efficiency (2.105 s vs. 18.711 s; pool scaling $\approx 10.4\times$ and $\approx 9.4\times$).

Token-decimal normalization, USD price oracles, and permit-versus-approve coverage may affect edge weights. Extended baselines (alternative PageRank variants and lending-oriented graphs) are archived for future comparison (Appendix 8.1) but are excluded from the main dissertation narrative. Unsecured lending default labels are out of scope. Regulatory and ethical deployment constraints (Section 5.6) limit how scores should be used even when alignment is favorable.

These limitations bound the claims: EndorseRank is a credible, efficient, allowance-specific social

reputation method on the studied cohort and window; it is not a universal credit score, a Sybil-proof identity system, or a replacement for protocol-level collateralization ^{[26],[48]}.